TEM-Analysis Repository Manual

CdSe/CdS Transmission Electron Microscopy (TEM) Image Analysis

Version 0.2 - Enright et al. 2018 reference dataset

1. Purpose of this project

This software analyzes grayscale TEM images of CdSe and CdS nanoparticles. Given an image with a visible scale bar, it:

1. Finds dark particle blobs on the carbon film background
2. Measures length and width in nanometers
3. Labels each particle (rod, dot, or reject - depending on analysis mode)
4. Saves an annotated overlay image and a CSV spreadsheet

Scientific reference: Enright, M. B. et al. *Mater. Chem. Front.* 2018, 10.1039/c8qm00056e. Supplementary Figure S2 provides validation TEM panels and published rod dimensions.

What this is not: A machine-learning black box. It uses classical computer vision (thresholding, shape metrics). It works best on clean TEM exports; screenshots and very dense clusters are harder.

2. Who this manual is for

Reader	Start here
Occasional user	Section 4 (three ways to run)
Someone reading the code	Section 6 (file guide)
Developer tuning parameters	Section 7 + docs/DEVELOPMENT.md

3. Quick start (10 minutes)

Install once

```
git clone https://github.com/fayh601-glitch/TEM-analysis.git
cd TEM-analysis
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
pip install -e .
```

Analyze with guided script

```
bash scripts/analyze_my_image.sh
```

Follow the prompts. Results go to outputs/user_runs/.

Read your results

Output file	Description
*_overlay.png	Image with colored particle outlines and size labels
*_measurements.csv	One row per detected particle
*_segments_debug.png	All segmented blobs numbered (troubleshooting)

4. Three ways to analyze an image

A. Guided shell script (easiest)

```
bash scripts/analyze_my_image.sh /path/to/image.png
```

Prompts for particle type, scale bar nm, and scale bar pixels.

B. Web upload (Streamlit)

```
pip install -r requirements-optional.txt
streamlit run app/streamlit_app.py
```

Opens a browser window. Upload image, enter scale bar, click Analyze.

C. Terminal command (full control)

```
tem-rods analyze \
  --image data/curated/s2_A_starting_rods.png \
  --preset enright_rods \
  --mode rods \
  --scale-bar-nm 20 \
  --scale-bar-pixels 48 \
  --output-dir outputs/my_run
```

5. Analysis modes

Use one codebase with a mode flag - not separate git branches.

Mode	CLI flag	Meaning
Both	--mode both	Classify rods and dots (default)

Presets bundle tuned settings:

Preset	Mode	Best for
enright_rods	rods	Enright SI Figure S2 nanorod panels
dots_only	dots	Spherical quantum dot samples
screenshot	both	Paper figure screenshots with white margins
sparse_cluster	both	Large sparse fields (e.g. 200 nm scale bar)

6. How the pipeline works

```
  TEM image (pixels)
        ?
        ?
  ???????????????
  ?  io.py      ?  Load PNG / TIFF / JPG as grayscale array
  ???????????????
        ?
        ?
  ???????????????
  ? calibrate   ?  scale_bar_nm ÷ scale_bar_pixels -> nm/pixel
  ???????????????
        ?
        ?
  ???????????????
  ? preprocess  ?  Normalize contrast, light blur (reduce grain)
  ???????????????
        ?
        ?
  ???????????????
  ? segment     ?  Threshold -> find dark blobs -> filter noise
  ???????????????      mask scale-bar strip, drop faint blobs
        ?
        ?
  ???????????????
  ? measure     ?  Ellipse fit -> length, width, eccentricity
  ???????????????
        ?
        ?
  ???????????????
  ? classify    ?  rod / dot / reject (+ analysis mode filter)
  ???????????????
        ?
        ?
  ???????????????
  ? pipeline    ?  Write CSV + overlay PNG + optional debug image
  ???????????????
```

Measurement definitions

Quantity	Definition
Length	Major axis of fitted ellipse (longest direction)
Width	Minor axis of fitted ellipse (shortest direction)
Rod	High eccentricity AND aspect ratio above threshold
Dot	Low eccentricity AND roughly round
Reject	Ambiguous shape, wrong mode, or filtered artifact

Overlay legend

Visual	Meaning
Solid green contour	Rod boundary
Dashed green ellipse	Fitted length/width axes
Blue	Dot (only in both mode)
Orange dashed	Reject - segmented but not reported as rod/dot
R 28.0×3.5 nm	Class letter + dimensions

Blank regions on the overlay mean the particle was never segmented (too faint, merged, or below area threshold). Compare with *_segments_debug.png.

7. Repository layout

```
TEM-analysis/
??? app/
?   ??? streamlit_app.py       Web upload UI
??? src/tem_rods/              Main Python package
?   ??? cli.py                 Terminal: tem-rods analyze
?   ??? pipeline.py            End-to-end workflow
?   ??? io.py                  Load images
?   ??? preprocess.py          Contrast + denoise
?   ??? segment.py             Find particles (most critical)
?   ??? measure.py             Length / width in nm
?   ??? classify.py            Rod / dot / reject + mode filter
?   ??? models.py              Settings and data types
?   ??? calibrate.py           Pixels -> nanometers
?   ??? scale_bar.py           Auto-detect scale bar
?   ??? presets.py             Tuned config bundles
??? scripts/
?   ??? analyze_my_image.sh    Guided analysis (beginners)
?   ??? run_demo.sh            Analyze 3 paper images
?   ??? extract_pdf_images.py  Pull figures from SI PDF
?   ??? prepare_paper_dataset.py
?   ??? build_manual_pdf.py    Regenerate this PDF
??? data/
?   ??? curated/               Enright S2 TEM panels
?   ??? calibration.csv        Scale bar per image
?   ??? labels/                Manual ground-truth labels
??? outputs/
?   ??? demo/                  Example results (on GitHub)
?   ??? user_runs/             Your analyses
??? tests/                     Automatic tests (pytest)
??? docs/
    ??? GETTING_STARTED.md     Beginner walkthrough
    ??? HOW_IT_WORKS.md        Plain-English pipeline guide
    ??? DEVELOPMENT.md         Contributor / tuning guide
    ??? PROJECT_STATUS.md      Roadmap
    ??? TEM-analysis-Manual.pdf  This document (PDF)
```

8. Source code file guide

Read files in this order if you are new to the code:

1. pipeline.py - Orchestrates everything; best single entry point
2. segment.py - Where most accuracy is won or lost
3. classify.py - Shape rules and analysis modes
4. measure.py - Converts blob geometry to nm
5. models.py - All tunable parameters in one place
6. cli.py - How Terminal flags map to settings

Each file has a comment block at the top explaining its job in plain English.

9. Key parameters (models.py / CLI)

Parameter	Typical value	Purpose
min_particle_area_px	80-150	Ignore tiny specks
min_eccentricity_rod	0.82-0.85	How elongated = rod
min_aspect_ratio_rod	1.38-1.5	Length/width ratio for rod
mask_bottom_fraction	0.10	Ignore scale-bar strip
split_touching_particles	off for Enright	Split merged clusters (can fragment rods)
analysis_mode	rods / dots / both	Sample-type filter

10. Troubleshooting

Problem	Likely cause	What to try
All sizes wrong	Bad scale bar pixels	Re-measure white line in ImageJ
Missing rods	Segmentation	Lower min_area; try screenshot preset; check debug PNG
False green rods	Noise / caption text	Crop to TEM field only; raise rod thresholds
Many orange rejects	Dense overlaps	Expected for clusters; tune segmentation
Blue dots on rod-only sample	Round fragments	Use --mode rods or enright_rods preset
Zero particles	Wrong preset/threshold	Try screenshot preset; check image is grayscale TEM

11. Validation data

Curated Enright SI panels in data/curated/:

File	Paper mean length × width
s2_A_starting_rods.png	27.6 × 2.9 nm
s2_B_30min.png	44.4 × 4.4 nm
s2_D_65min.png	99.3 × 5.9 nm

Run demo: bash scripts/run_demo.sh
Compare: outputs/validation/paper_comparison.csv

12. Testing

```
pytest
```

Tests use synthetic images with known rod and dot shapes to catch regressions.

13. Roadmap (known limitations)

1. Dense touching rods merge - segmentation is the main bottleneck
2. Screenshot images perform worse than raw TEM exports
3. UV-Vis correlation not yet implemented
4. Hand labels in data/labels/manual_v1.csv needed for rigorous validation

14. Commands reference

```
# Analyze one image (rods only)
tem-rods analyze -i IMAGE.png --preset enright_rods --mode rods \
  --scale-bar-nm 20 --scale-bar-pixels 45 -o outputs/my_run

# Auto scale bar
tem-rods analyze -i IMAGE.png --auto-scale-bar --scale-bar-nm 20 --preset enright_rods

# Demo all paper panels
bash scripts/run_demo.sh

# Web UI
streamlit run app/streamlit_app.py

# Regenerate manual PDF
python scripts/build_manual_pdf.py
```

15. License and citation

Academic / research use. TEM reference images derived from Enright et al. 2018 supplementary information.

When publishing results obtained with this tool, cite the Enright paper for validation methodology and document your scale-bar calibration and analysis parameters.

End of manual